

Development of assessment framework for environmental sustainability in higher education institutions

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framework

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Abstract

Purpose – Unsustainable actions have resulted in the deterioration of the environment. Higher education institutions are social entities responsible for developing future leaders who have the skills and abilities to solve societal challenges. This study aims to develop a conceptual assessment model of environmental sustainability in higher education and apply the model to test the environmental sustainability level of a higher education institution.

Design/methodology/approach – The fuzzy logic method is used to develop the conceptual model, and a higher education institution in India was selected for evaluating the environmental sustainability level.

Findings – The fuzzy environmental sustainability index for the higher education institution selected for the study was computed as (3.58, 4.95, 6.33). The computed value identifies the institution as “environmentally sustainable”. The calculation of the fuzzy performance importance index further helped in identifying weak attributes that need improvement.

Research limitations/implications – The result of this study would help the management understand their current environmental sustainability level and take corrective measures to improve the weak attributes. It would also assist the institution in reworking its processes and practices.

Originality/value – This study has contributed to the literature of environmental sustainability assessment by creating a conceptual model for evaluating environmental sustainability in higher education. This model can be applied to any higher education institution.

Keywords Environmental sustainability, Assessment, Higher education institutions, Decision support framework, Fuzzy logic

Paper type Research paper

1. Introduction

Unsustainable production and consumption patterns have resulted in environmental degradation as well as rising economic and social costs (United Nations, Department of Economic and Social Affairs, 2013). It was the Brundtland (1987) report that sparked global recognition and concern for environmentally sustainable, economically viable and socially just growth and development. The report underlined that development should meet the current generation’s requirements without affecting future generations’ ability to meet their demands. Education has the potential to raise public consciousness about the implications and ramifications of unsustainable behaviour on society (UN DESD, 2005). Educational institutions, in general, and higher education (HE) in particular, can play a crucial role in creating awareness among the present generation. HE institutions (HEIs) serve as real-life laboratories, where students learn from their experiences, acquire knowledge and apply this knowledge in solving societal problems (Velazquez *et al.*, 2006). In pursuit of imparting environmental education, several HEIs have signed declarations and have either



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restructured or are in the process of restructuring their programmes and practices (UNESCO, 2005; Wals, 2014). To help students understand the real-world problems and reflect on their values, beliefs and actions (Chhokar, 2010; Sriskandarajah *et al.*, 2010), HEIs need to incorporate sustainability principles into the curriculum, operations, community outreach programmes and research (Menon and Suresh, 2020). Integrating sustainability across activities should follow the assessment and reporting of sustainability initiatives and progress (Ceulemans *et al.*, 2015). Several frameworks and assessment tools are available for assessing the sustainability of HEIs. Some exclusively focus on the environmental dimension, while others are more comprehensive in approach, with equal impetus on all three sustainable dimensions (environment, economic and equity). The lack of understanding of the sustainability concept and complex procedures have discouraged HEIs from using these tools (Alba-Hidalgo *et al.*, 2018). According to Yarime and Tanaka (2012), HEIs should adopt multiple tools and methods for assessing sustainability. None of the studies so far has applied the fuzzy logic method to measure the environmental sustainability in HE, and this became the motivation for developing an assessment framework using fuzzy logic. This study attempts to develop an assessment framework that is easy to understand and allows for quantitative measurement of environmentally sustainable initiatives in HE. The following research objectives were formulated:

- to comprehend the enablers, criteria and attributes of environmental sustainability and propose a conceptual assessment framework for HEIs;
- to measure the environmental sustainability level of a HEI using the fuzzy logic method; and
- to address the weak attributes for case HEI and propose suggestions for overcoming the impediments.

2. Literature review

2.1 *Environmental sustainability in higher education*

The first step towards a green campus begins with green curricula. A green curriculum focuses on incorporating environmental content into the academic programme in the form of a case study on environmental issues, as a module, an elective course or as a programme on environmental sustainability (Chhokar, 2010; Sammalisto and Lindhqvist, 2008; Wu *et al.*, 2015). The depth and the extent of environmental education integration into the curriculum vary across disciplines and institutions (Pavlova, 2013). Finlay and Massey (2012) reiterated institutionalisation of sustainability is crucial for imparting environmental awareness among students. Some of the sustainable practices adopted by the HEIs include organic botanical gardens (Cheang *et al.*, 2017), green energy (Bull *et al.*, 2018), water management (Mohammadalizadehkorde and Weaver, 2018), solid waste management (Paghasian, 2017), green building (Moh'd Khier Momani *et al.*, 2020), recycling awareness campaigns (Krizek *et al.*, 2012), conservation of natural resources (Murray, 2018), sustainable labs (Wright *et al.*, 2008) and environment/sustainable management systems (Amaral *et al.*, 2015) for monitoring carbon footprints on campuses. The university–community joint research projects aiming at conserving the environment and natural resources of the local community is gradually becoming prevalent in colleges (Brundiars *et al.*, 2010). Studies have identified government policies and programmes, leadership, institutional commitment, accreditation and sustainability audits as factors that can catalyse the integration of sustainability in HE (Nicholls *et al.*, 2013; Hargreaves, 2007).

2.2 Assessment of environmental sustainability

The fifth and the most important dimension of sustainability in HE is assessment and reporting, the other four dimensions being teaching, research, sustainable practices, and outreach (Lozano, 2006). Regular monitoring, measuring, and reporting are crucial for any HEI committed to promoting education for sustainability (Ceulemans *et al.*, 2015). Several frameworks and assessment techniques are available for analysing and assessing the sustainability level in HEIs (Alshuwaikh and Abubakar, 2008; Shriberg, 2002; Lambrechts and Ceulemans, 2013; Berzosa *et al.*, 2017; Lauder *et al.*, 2015). Despite several tools available, the application of these tools in evaluating sustainability in HE is varied. The lack of specificity, lack of quantitative evaluation, the inclusion of several criteria and complicated procedure have discouraged HEIs from using these tools (Du *et al.*, 2020). Although holistic assessment tools are commonly promoted (Disterheft *et al.*, 2015), this study follows a reductionist approach, focusing on developing a comprehensive environmental sustainability assessment framework that is easy to understand and allows for quantitative measurement.

3. Research methodology

This study uses the fuzzy logic method to assess the environmental sustainability level and identify the weak attributes in the selected HEI. The fuzzy logic method was introduced by Zadeh (1965). The method helps deal with ambiguous, imprecise, unclear phenomena. It is an effective method for resolving multiple criteria conflicts and for providing better evaluating solutions (Singh *et al.*, 2013). It enables data collection and communication of results in linguistic terms. Linguistic terms, such as “really bad” and “worst”, are difficult to translate and have a slight variation in meaning. Fuzzy logic helps convert linguistic terms to numerical values. The fuzzy intervals, also known as membership functions, rule out the compatibility and consistency challenges of converting linguistic phrases to numbers. Researchers have applied the fuzzy logic method in manufacturing and service sectors alike. Studies by Lin and Chen (2004), Dahmardeh and Pourshahabi (2011), Lin *et al.* (2006), Shahrabi (2011), Lotfi and Houshmand (2014), Agrawal *et al.* (2017) and Vinodh *et al.* (2012) applied fuzzy logic to measure the agility of the manufacturing organisations. Vinodh and Aravindraj (2015) used a 40-criteria agility assessment model and applied multi-grade and fuzzy logic approaches to benchmark their results. Researchers have used fuzzy logic to understand the implementation of lean practices in health care (Narayanamurthy and Gurumurthy, 2018). The literature revealed limited research on the use of fuzzy logic in assessing sustainability. Vimal *et al.* (2015) used the multi-grade fuzzy method to evaluate the process sustainability of a manufacturing firm. Rajak and Vinodh (2015) applied the fuzzy method to identify the social sustainability indicators and measure the sustainability performance of an automotive component manufacturing organisation. The notations used for fuzzy logic calculations in this study are shown in Table 1 (Figure 1).

3.1 Sampling design and data collection

A thorough literature review and inputs from academicians helped develop the conceptual model for environmental sustainability assessment in HE. This was followed by an interview with five senior faculty members of HEIs involved in supporting environmental sustainability activities in their respective institutions. The faculty members were first sent a mail, and after receiving their consent to participate in the research, they were contacted for collecting importance weights for the environmental sustainability enablers, criteria and attributes. The next step was to interview faculty members at the HEI chosen for evaluation. The IQAC and NSS coordinators, as well as three faculty members, were among the

Table 1.
Notations used for
fuzzy logic
environmental
sustainability
assessment model for
HEI

Indices	Abbreviations
R_i	Fuzzy performance rating for environmental sustainability of HEI of i^{th} enabler
R_{ij}	Fuzzy performance rating for environmental sustainability of HEI of j^{th} criterion in i^{th} enabler
R_{ijk}	Fuzzy performance rating for environmental sustainability of HEI of k^{th} attribute of j^{th} criterion in i^{th} enabler
G_i	Fuzzy importance weight for environmental sustainability of HEI of i^{th} enabler
G_{ij}	Fuzzy importance weight for environmental sustainability of HEI of j^{th} criterion in i^{th} enabler
G_{ijk}	Fuzzy importance weight for environmental sustainability of HEI of k^{th} attribute of j^{th} criterion in i^{th} enabler
ES_i	Environmental sustainability measure of i^{th} enabler
ES_{ij}	Environmental sustainability measure of j^{th} criterion in i^{th} enabler
ES_{ijk}	Environmental sustainability measure of k^{th} attribute of j^{th} criterion in i^{th} enabler
$FESI$	Fuzzy environmental sustainability index of HEI
$FESL_i$	Fuzzy number of natural language expression set of environmentally sustainable level
$fFESL_i(x)$	Triangular fuzzy number of $FESL_i$
$fFESL_i(x)$	Triangular fuzzy number of $FESL_i$

participants. The faculty members were selected based on their involvement in environmental sustainability practices on- and off-campus. The information was gathered using a questionnaire and a scheduled interview. Each respondent was interviewed for 30–40 min. In this area of research, a sample of five respondents is sufficient to achieve the research objectives (Vinodh and Balaji, 2011).

4. Conceptual model

The assessment was executed in two steps: the first step involved developing a conceptual assessment model of environmental sustainability in HE (Vinodh and Vimal, 2012; Vinodh and Aravindraj, 2015), and the second step involved testing the model in a case HEI. The conceptual model is divided into three levels: enablers, criteria and attributes. The conceptual model with three enablers, six criteria and 35 attributes is comprehensive to evaluate the environmental sustainability level of HEIs. A thorough literature review and discussions with the academic experts helped identify the attributes that can facilitate environmental sustainability in HE. The conceptual model is presented in Table 2.

4.1 About the higher education institution selected for the assessment

The institution selected is a 65-year-old college situated in India. The college has been honoured with several awards and recognitions for its contribution towards conserving the environment. It is the recipient of Bhoomimithra award, Palathully Award, National Environmental Awareness Award, the Green Award and “One District One Green” award conferred by Mahatma Gandhi National Council of Rural Education (MGNCRE). The institution was interested in learning about its environmental sustainability level, which is why this college was chosen for the evaluation.

4.2 The assessment procedure using the fuzzy logic method

Identification of the enablers, criteria and environmental sustainability attributes is followed by the evaluation procedure presented below:

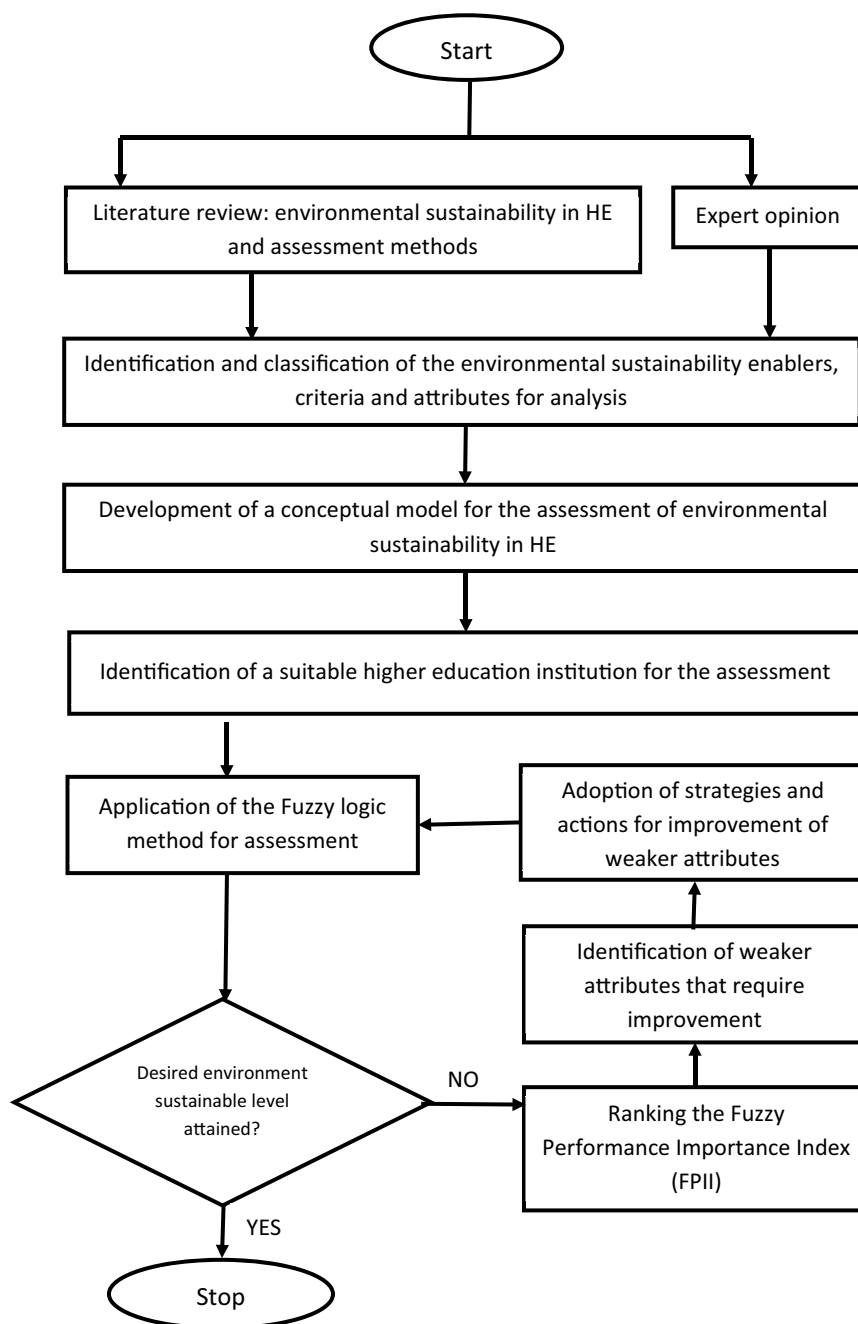


Figure 1.
Schematic
representation of the
research
methodology

Table 2.
Environmental
sustainability
assessment model for
HEI

Enabler	Criteria	Attributes	References
Management (<i>ES₁</i>)	Infrastructure (<i>ES₁₁</i>)	Use of solar panels as an alternative source of energy (<i>ES₁₁₁</i>) Purchase of energy-efficient appliances and instruments (<i>ES₁₁₂</i>) Construction of campus building that ensures proper ventilation and allows for natural sunlight in classrooms (<i>ES₁₁₃</i>) Rainwater harvesting facility (<i>ES₁₁₄</i>) Investment in reverse osmosis plant (<i>ES₁₁₅</i>) Investment in sensor-based lighting (<i>ES₁₁₆</i>) Solid waste management system (<i>ES₁₁₇</i>)	Moh'd Khier Momani <i>et al.</i> (2020) Geng <i>et al.</i> (2013) Moh'd Khier Momani <i>et al.</i> (2020), de Castro and Jabbour (2013) Keskar <i>et al.</i> (2016) Jiang <i>et al.</i> (2018), Hamzah <i>et al.</i> (2018) Liang <i>et al.</i> (2013), Riyanto <i>et al.</i> (2018) Geng <i>et al.</i> (2013), de Castro and Jabbour (2013); Espinosa <i>et al.</i> (2008) Oyedepo <i>et al.</i> (2019) Dagiliūtė and Liobikienė (2015) Nicholls <i>et al.</i> (2013)
	Institutional commitment (<i>ES₁₂</i>)	Regular energy and environmental audits (<i>ES₁₂₁</i>) Funding environmental sustainability research projects (<i>ES₁₂₂</i>) Complying with environmental sustainability requirements of accreditation bodies (<i>ES₁₂₃</i>) Reuse of water from labs in watering plants on campus (<i>ES₁₂₄</i>) Provision for composting food waste (<i>ES₁₂₅</i>) Organic gardens/farming (<i>ES₁₂₆</i>) Planting trees around the campus to maintain lower temperature and reduce carbon footprint (<i>ES₁₂₇</i>) Landscaping using plants that consume less water (<i>ES₁₂₈</i>) Provision for biodegradable and non-biodegradable waste management and disposal (<i>ES₁₂₉</i>) Ban of single-use plastic (<i>ES₁₂₁₀</i>) Reduce use of paper on campus (<i>ES₁₂₁₁</i>) Repair and reuse of electronic gadgets (<i>ES₁₂₁₂</i>) Mechanism for e-waste management and disposal (<i>ES₁₂₁₃</i>)	Jayashree <i>et al.</i> (2014) Waliczek <i>et al.</i> (2016) Cheang <i>et al.</i> (2017) Moh'd Khier Momani <i>et al.</i> (2020) Zorn and Panton (2016) Paghasian (2017) UGC (2019), Choate <i>et al.</i> (2018) Kapuka <i>et al.</i> (2017) Saldaña-Durán and Messina-Fernández (2020) Saldaña-Durán and Messina-Fernández (2020); Odhiambo (2009) Hamzah <i>et al.</i> (2018) Moh'd Khier Momani <i>et al.</i> (2020), Singhal (2012) Yanthi <i>et al.</i> (2019)
Faculty members and students (<i>ES₂</i>)	Creating awareness (<i>ES₂₁</i>)	Awareness campaigns by faculty members and students (<i>ES₂₁₁</i>) Organising seminars and student-driven summits (<i>ES₂₁₂</i>) Use of stickers and signs around campus as reminders (<i>ES₂₁₃</i>)	

(continued)

Enabler	Criteria	Attributes	References
External environment (<i>ES₃</i>)	Sustainable practices (<i>ES₂₂</i>)	Use of public transportation by students and staff (<i>ES_{22.1}</i>) Use of carpooling/eco-friendly vehicles by staff/students (<i>ES_{22.2}</i>) Unplugging electronic gadgets temporarily not in use (<i>ES_{22.3}</i>) Use of hand sanitisers in common places to reduce the use of water (<i>ES_{22.4}</i>) Use of reusable utensils on campus (<i>ES_{22.5}</i>) Faculty/student involvement in sustainability related research (<i>ES_{22.6}</i>) Educating neighbouring community on sustainable practices (<i>ES_{22.7}</i>) Incorporating environmental sustainability principles and values into the curriculum (<i>ES_{31.1}</i>) Student participation in community engagements (<i>ES_{31.2}</i>) Assigning more weightage to sustainable actions on campus to facilitate the integration of sustainability initiatives across activities (<i>ES_{31.3}</i>) Funding sustainable project (<i>ES_{32.1}</i>)	Moh'd Khier Momani <i>et al.</i> (2020); Freidenfelds <i>et al.</i> (2018) Luo <i>et al.</i> (2020) Leal Filho <i>et al.</i> (2019) Fournier and Berry (2012) UGC (2019) Singhal (2012), Menon and Suresh (2021) Jadhav <i>et al.</i> (2014) Jadhav <i>et al.</i> (2014), Hamzah <i>et al.</i> (2018); Kabera (2017) UGC (2019), Singh (2018) NAAC (2019)
	Accreditation agencies (<i>ES₃₁</i>)	Rewarding and recognising colleges for their contribution towards environmental sustainability (<i>ES_{32.2}</i>)	Parvez and Agrawal (2019); Dagiliūtė and Liobikienė (2015) Filho <i>et al.</i> (2015)
	Government support (<i>ES₃₂</i>)		

Table 2.

- Determination of an appropriate linguistic scale for assessing importance weightings and performance ratings of environmental sustainability attributes.
- Evaluation of importance weights and performance ratings of environmental sustainability attributes.
- Approximation of linguistic expressions to a set of triangular fuzzy numbers.
- Determination of fuzzy environmental sustainability index (FESI).
- Determination of Euclidean distance to match FESI with appropriate environmental sustainability level.
- Identification and analysis of obstacles for improvement.

Step 1: Determination of an appropriate linguistic scale for assessing importance weighting and performance rating of environmental sustainability attributes. Linguistic terms help capture responses from experts in situations where it is difficult to determine the score of a vague attribute. The linguistic scale used in this study is adopted from the previous studies by [Lin et al. \(2006\)](#) and [Vinodh and Devadasan \(2011\)](#). The scale to assess performance ratings of environmental sustainability capabilities includes the following linguistic variables: worst (W), very poor (VP), poor (P), fair (F), good (G), very good (VG), excellent (E) and Likewise, linguistic variables used to assess the importance weights are: very low (VL), low (L), fairly low (FL), medium (M), fairly high (FH), high (H) and very high (VH).

The sample questions are given below:

Please tick against an appropriate option depending on how important you think are the enablers, criteria and attributes in the assessment of environmental sustainability in HE.

(1) Use of solar panels as an alternate source of energy

Very low () low () Fairly low () Medium () Fairly high () High () Very high ()

Please rate the performance of the environmental sustainability attribute in your institute:

(1) Use of solar panels as an alternate source of energy

Worst () Very poor () Poor () Fair () Good () Very good () Excellent ()

Step 2: Evaluation of importance weights and performance ratings of environmental sustainability attributes. At this stage, academic experts are contacted for collecting importance weights and performance ratings for environmental sustainability attributes. Importance weights are collected for all the enablers, criteria and attributes, whereas performance ratings are collected for attributes alone. As fuzzy analysis enables aggregating performance attribute rating to criteria rating, and criteria rating to enablers rating, only attributes are considered for rating performance. The importance weights and performance ratings are aggregated using the average operational method, a robust method for a small sample.

Formula of average operational method =

$$(a_1b_1c_1)+,\dots,\dots,\dots,(a_nb_nc_n) = [(a_1+,\dots,\dots,+a_n)/n,(b_1+,\dots,\dots,+b_n)/n,(c_1+,\dots,\dots,+c_n)/n]$$

[Tables 3, 4 and 5](#) exhibit the importance weights and performance ratings collected from the respondents.

Step 3. Approximation of linguistic terms to a set of triangular fuzzy numbers. A group of fuzzy boundaries is referred to as a fuzzy set. Defined fuzzy classes for each variable allows for intermediate levels of membership ([Thakre et al., 2017](#)). The fuzzy numbers allow for transition from linguistic terms to numerical terms enabling simple calculations. The

performance rating scale ranges between 0 and 10, and the importance weight scale ranges between 0 and 1. Each linguistic variable is matched to a set of fuzzy numbers (a, b, c) , also known as the membership function ($\mu_A(X)$), shown in Table 6.

$$f_A(x) = \begin{cases} \frac{x-a}{b-a}, & a \leq x \leq b \\ \frac{x-c}{c-b}, & b \leq x \leq c \\ 0, & \text{otherwise} \end{cases}$$

Step 4: Determination of FESI. FESI represents the overall environmental sustainability level of the institute. To obtain the FESI, firstly, the fuzzy environmental sustainability of criteria (R_{ij}) is calculated using equation (1). Using equation (2), the environmental sustainability of the enabler (R_i) is calculated, and lastly, the overall FESI of the institute is obtained using equation (3).

$$R_{ij} = \frac{\sum_{k=1}^k (G_{ijk} \times R_{ijk})}{\sum_{k=1}^k G_{ijk}} \quad (1)$$

$$R_i = \frac{\sum_{j=1}^j (G_{ij} \times R_{ij})}{\sum_{j=1}^j G_{ij}} \quad (2)$$

$$FESI = \frac{\sum_{i=1}^i (G_i \times R_i)}{\sum_{i=1}^i G_i} \quad (3)$$

Sample computation of FESI

[Enabler	E1	E2	E3	E4	E5	Table 3. Importance weights of enablers of environmental sustainability of HEI
ES_1	H	M	H	H	H	
ES_2	VH	M	FH	FH	FH	
ES_3	FH	L	H	FH	H	

Criteria	E1	E2	E3	E4	E5	Table 4. Importance weights of criteria of environmental sustainability of HEI
ES_{11}	FH	L	H	H	VH	
ES_{12}	H	M	H	VH	H	
ES_{21}	FH	L	H	H	H	
ES_{22}	VH	H	H	VH	H	
ES_{31}	FH	L	FH	FH	M	
ES_{32}	FH	M	H	H	H	

Table 5.
Importance weights
and performance
rating of
environmental
sustainability of HEI

Attributes	Weights					Ratings				
	E1	E2	E3	E4	E5	R1	R2	R3	R4	R5
ES ₁₁₁	H	L	FL	H	H	F	P	P	P	F
ES ₁₁₂	M	L	M	FH	H	G	F	P	F	VP
ES ₁₁₃	VH	VL	H	H	FH	VG	G	F	F	P
ES ₁₁₄	FH	M	H	H	FH	E	G	G	E	VG
ES ₁₁₅	H	FH	VH	H	H	G	P	F	G	P
ES ₁₁₆	M	FL	H	FL	H	P	P	VP	VP	W
ES ₁₁₇	M	L	VH	M	FH	E	P	P	G	W
ES ₁₂₁	FH	L	FH	VH	FH	G	F	VP	VP	F
ES ₁₂₂	FH	FL	VH	FL	L	F	F	VP	P	VP
ES ₁₂₃	H	FL	H	FH	FH	G	G	VP	P	F
ES ₁₂₄	FH	FL	FL	FH	H	G	W	F	F	W
ES ₁₂₅	M	L	M	FL	FH	E	G	VP	F	W
ES ₁₂₆	FL	FL	FH	FH	FH	G	F	VP	F	W
ES ₁₂₇	FL	FH	H	H	FH	E	E	F	G	VG
ES ₁₂₈	L	L	H	FL	FH	F	G	F	G	VG
ES ₁₂₉	L	L	VH	H	H	E	VG	F	G	P
ES ₁₂₁₀	H	VH	H	H	H	VG	G	F	F	G
ES ₁₂₁₁	FH	M	H	H	H	P	P	F	VP	F
ES ₁₂₁₂	FL	M	FH	FH	FH	G	VG	VP	P	W
ES ₁₂₁₃	M	L	H	FH	FH	F	F	W	P	P
ES ₂₁₁	L	L	VH	H	M	E	F	VP	G	F
ES ₂₁₂	M	FH	H	H	M	G	F	P	F	F
ES ₂₁₃	FL	L	H	M	FH	E	F	VP	G	P
ES ₂₂₁	M	H	VH	H	M	VP	W	W	W	W
ES ₂₂₂	M	M	H	M	L	F	P	W	F	P
ES ₂₂₃	M	L	VH	H	H	VG	F	F	F	F
ES ₂₂₄	FH	FL	FH	FH	H	E	E	F	VG	G
ES ₂₂₅	L	L	H	FH	H	G	F	W	P	VP
ES ₂₂₆	FH	M	H	M	M	VG	G	F	VG	W
ES ₂₂₇	FL	L	M	M	M	F	F	G	F	P
ES ₃₁₁	FH	M	VH	VH	M	VG	F	G	F	G
ES ₃₁₂	FH	M	FH	H	M	F	P	VP	P	P
ES ₃₁₃	FH	FL	FH	M	M	E	F	VP	F	W
ES ₃₂₁	H	M	H	H	H	G	E	VP	P	W
ES ₃₂₃	FH	FL	VH	H	H	VG	E	VG	P	P

Table 6.
Linguistic terms and
appropriate fuzzy
numbers for ratings
and weights for
environmental
sustainability
assessment of HEI

Performance ratings		Importance weights	
Linguistic variable	Fuzzy no.	Linguistic variable	Linguistic variable
Worst (W)	(0,0.5,1.5)	Very low (VL)	(0,0.05,0.15)
Very Poor (VP)	(1,2,3)	Low (L)	(0.1,0.2,0.3)
Poor (P)	(2,3.5,5)	Fairly low (FL)	(0.2,0.35,0.5)
Fair (F)	(3.5,7)	Medium (M)	(0.3,0.5,0.7)
Good (G)	(5,6.5,8)	Fairly high (FH)	(0.5,0.65,0.8)
Very good (VG)	(7,8,9)	High (H)	(0.7,0.8,0.9)
Excellent (E)	(8.5,9.5,10)	Very high (VH)	(0.85,0.95,1.0)

The fuzzy interval values are assigned to importance weights and performance ratings using [Table 6](#). The aggregated importance weights and performance ratings are calculated based on the average operational method.

Say for attribute: ES_{111} (use of solar panels as alternative source of energy):

$$\begin{aligned}\text{Average fuzzy importance weights} &= [H + L + FL + H + H]/5 \\ &= [(0.7, 0.8, 0.9) + (0.1, 0.2, 0.3) + (0.2, 0.35, 0.5) + (0.7, 0.8, 0.9) + (0.7, 0.8, 0.9)]/5 \\ &= (0.48, 0.59, 0.7)\end{aligned}$$

$$\begin{aligned}\text{Average fuzzy performance ratings} &= [F + P + P + P + F]/5 \\ &= [(3, 5, 7) + (2, 3.5, 5) + (2, 3.5, 5) + (2, 3.5, 5) + (3, 5, 7)]/5 \\ &= (2.4, 4.1, 5.8)\end{aligned}$$

The next step is the calculation of fuzzy index rating for criteria (R_{ij}), which is done using [equation \(1\)](#). For example, the criterion rating for infrastructure (ES_{11}) is computed as given below:

$$\begin{aligned}ES_{11} &= [((2.4, 4.1, 5.8) \otimes (0.48, 0.59, 0.7)) \oplus ((2.8, 4.4, 6) \otimes (0.38, 0.53, 0.68)) \oplus \\ &\quad ((4, 5.6, 7.2) \otimes (0.55, 0.65, 0.75)) \oplus ((6.8, 8, 9) \otimes (0.54, 0.68, 0.82)) \oplus \\ &\quad ((3.4, 5, 6.6) \otimes (0.69, 0.8, 0.9)) \oplus ((1.2, 2.3, 3.5) \otimes (0.42, 0.56, 0.7)) \oplus \\ &\quad ((3.5, 4.7, 5.9) \otimes (0.41, 0.56, 0.7))] \\ &= \frac{[(0.48, 0.59, 0.7) \oplus (0.38, 0.53, 0.68) \oplus (0.55, 0.65, 0.75) \oplus (0.54, 0.68, 0.82) \oplus \\ &\quad (0.69, 0.8, 0.9) \oplus (0.42, 0.56, 0.7) \oplus (0.41, 0.56, 0.7)]}{ES_{11} = (3.565, 4.977, 6.369)}\end{aligned}$$

The same equation is used for computing fuzzy index values for all the other criteria. The fuzzy index of the enabler (ES_1) is calculated using [equation \(2\)](#). The calculation is shown below:

$$\begin{aligned}ES_1 &= \frac{[(3.565, 4.977, 6.369) \otimes (0.57, 0.68, 0.78)] \oplus [(3.457, 4.85, 6.261) \otimes (0.65, 0.77, 0.88)]}{[(0.57, 0.68, 0.78) \oplus (0.65, 0.77, 0.88)]} \\ ES_1 &= (3.507, 4.909, 6.312)\end{aligned}$$

The computed values of all the criteria and the enablers are in [Tables 7](#) and [8](#). The FESI is calculated using [equation \(3\)](#):

Table 7.
Fuzzy index of
environmental
sustainability of HEI
of criteria rating

Environmental sustainability criteria	Environmental sustainability attributes	Fuzzy performance average rating (R_{ijk})	Attributes importance average weight (G_{ijk})	Criteria rating
ES_{11}	ES_{111}	(2.4,4.1,5.8)	(0.48,0.59,0.7)	(3.565,4.977,6.369)
	ES_{112}	(2.8,4.4,6)	(0.38,0.53,0.68)	
	ES_{113}	(4.5,6.7,2)	(0.55,0.65,0.75)	
	ES_{114}	(6.8,8,9)	(0.54,0.68,0.82)	
	ES_{115}	(3.4,5,6.6)	(0.69,0.8,0.9)	
	ES_{116}	(1.2,2.3,3.5)	(0.42,0.56,0.7)	
ES_{12}	ES_{117}	(3.5,4.7,5.9)	(0.41,0.56,0.7)	(3.457,4.85,6.261)
	ES_{121}	(2.6,4.1,5.6)	(0.49,0.62,0.74)	
	ES_{122}	(2,3,5,5)	(0.37,0.5,0.62)	
	ES_{123}	(3.2,4.7,6.2)	(0.52,0.65,0.78)	
	ES_{124}	(2.2,3,5,5)	(0.42,0.56,0.7)	
	ES_{125}	(3.5,4.7,5.9)	(0.28,0.44,0.6)	
	ES_{126}	(2.4,3.8,5.3)	(0.38,0.53,0.68)	
	ES_{127}	(6.4,7.7,8.8)	(0.52,0.65,0.78)	
	ES_{128}	(4.6,6.2,7.8)	(0.32,0.44,0.56)	
	ES_{129}	(5.1,6.5,7.8)	(0.49,0.59,0.68)	
	ES_{1210}	(4.6,6.2,7.8)	(0.73,0.83,0.92)	
	ES_{1211}	(2.2,3.8,5.4)	(0.58,0.71,0.84)	
	ES_{1212}	(3.4,1,5,3)	(0.4,0.56,0.72)	
	ES_{1213}	(2,3,5,5,1)	(0.42,0.56,0.7)	
ES_{21}	ES_{211}	(4.1,5,6,7)	(0.41,0.53,0.64)	(3.688,5.278,6.8)
	ES_{212}	(3.2,5,6,8)	(0.5,0.65,0.8)	
	ES_{213}	(3.9,5,3,6,6)	(0.36,0.5,0.64)	
ES_{22}	ES_{221}	(0.2,0.8,1.8)	(0.57,0.71,0.84)	(3.14,4.474,5.902)
	ES_{222}	(2,3,5,5,1)	(0.34,0.5,0.66)	
	ES_{223}	(3.8,5,6,7,4)	(0.53,0.65,0.76)	
	ES_{224}	(6.4,7.7,8.8)	(0.48,0.62,0.76)	
	ES_{225}	(2.2,3,5,4,9)	(0.42,0.53,0.64)	
	ES_{226}	(4.4,5,6,6,9)	(0.42,0.59,0.76)	
	ES_{227}	(3.2,5,6,8)	(0.24,0.41,0.58)	
	ES_{311}	(4.6,6.2,7.8)	(0.56,0.71,0.84)	(3.342,4.787,6.225)
ES_{31}	ES_{312}	(2,3,5,5)	(0.46,0.62,0.78)	
	ES_{313}	(3.1,4.4,5.7)	(0.36,0.53,0.7)	
	ES_{321}	(3.3,4.4,5.5)	(0.62,0.74,0.86)	(4.275,5.428,6.525)
ES_{32}	ES_{323}	(5.3,6.5,7.6)	(0.59,0.71,0.82)	

Table 8.
Fuzzy index of
environmental
sustainability of HEI
of enabler rating and
FESI

Enabler	Criteria	Criteria rating (R_{ij})	Criteria weights (G_{ij})	Enabler rating (R_i)	Enabler weights (G_i)	Fuzzy environmental sustainability of HEI index ($FESI$)
ES_1	ES_{11}	(3.56,4.97,6.36)	(0.57,0.68,0.78)	(3.50,4.9,6.31)	(0.62,0.74,0.86)	(3.58, 4.95, 6.33)
	ES_{12}	(3.45,4.85,6.26)	(0.65,0.77,0.88)			
ES_2	ES_{21}	(3.68,5.27,6.8)	(0.54,0.65,0.76)	(3.37,4.82,6.3)	(0.53,0.68,0.82)	
	ES_{22}	(3.14,4.47,5.9)	(0.76,0.86,0.94)			
ES_3	ES_{31}	(3.34,4.78,6.22)	(0.38,0.53,0.68)	(3.90,5.15,6.39)	(0.5,0.62,0.74)	
	ES_{32}	(4.27,5.42,6.52)	(0.58,0.71,0.84)			

$$\begin{aligned} \text{FESI} = & [((3.507, 4.909, 6.312) \otimes (0.62, 0.74, 0.86)) \oplus ((3.368, 4.82, 6.3) \otimes \\ & (0.53, 0.68, 0.82)) \oplus ((3.9, 5.154, 6.39) \otimes (0.5, 0.62, 0.74))] \\ & [(0.62, 0.74, 0.86) \oplus (0.53, 0.68, 0.82) \oplus (0.5, 0.62, 0.74)] \end{aligned}$$

$$\text{FESI} = (3.583, 4.954, 6.333)$$

Step 5: Determination of Euclidean distance to match FESI with an appropriate environmental sustainability level. The Euclidean distance method, successive approximation and piecewise decomposition are the three commonly used techniques for matching membership functions with linguistic terms. This study has used the Euclidean distance method as it is the most insightful method for human proximity estimation (Vinodh *et al.*, 2013). The Euclidean distance is calculated using equation (4):

$$D(\text{FESI}, \text{FESL}_i) = \sqrt{\sum (f\text{FESI}(x) - f\text{FESL}_i(x))^2} \quad (4)$$

$$D(\text{FESI}, \text{EES}) = \left[(3.58 - 7)^2 + (4.95 - 8.5)^2 + (6.33 - 10)^2 \right]^{1/2} = 6.14$$

$$D(\text{FESI}, \text{VES}) = \left[(3.58 - 5.5)^2 + (4.95 - 7)^2 + (6.33 - 8.5)^2 \right]^{1/2} = 3.54$$

$$D(\text{FESI}, \text{ES}) = \left[(3.58 - 3.5)^2 + (4.95 - 5)^2 + (6.33 - 6.5)^2 \right]^{1/2} = 0.19$$

$$D(\text{FESI}, \text{FES}) = \left[(3.58 - 1.5)^2 + (4.95 - 3)^2 + (6.33 - 4.5)^2 \right]^{1/2} = 3.34$$

$$D(\text{FESI}, \text{SES}) = \left[(3.58 - 0)^2 + (4.95 - 1.5)^2 + (6.33 - 3)^2 \right]^{1/2} = 5.99$$

Natural language expression set of environmentally sustainable labels for HEI (<i>FESL_i</i>)	Fuzzy intervals
Extremely environmentally sustainable (EES)	(7, 8.5, 10)
Very environmentally sustainable (VES)	(5.5, 7, 8.5)
Environmentally sustainable (ES)	(3.5, 5, 6.5)
Fairly environmentally sustainable (FES)	(1.5, 3, 4.5)
Slowly environmentally sustainable (SES)	(0, 1.5, 3)

Table 9.
Environmental
sustainability level
for HEI and fuzzy
intervals

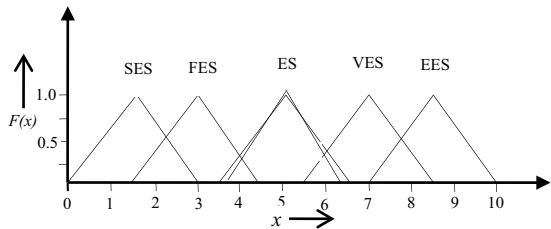


Figure 2.
Linguistic levels to
match FESI of HEI

Table 10.
FPII of
Environmental
sustainability of HEI

Environmental sustainability attributes	Fuzzy performance average rating (R_{ijk})	$U_{ijk} = (1, 1, 1)-G_{ijk}$	FPII	Ranking score
ES_{111}	(2,4,4,1,5,8)	(0.52,0.41,0.3)	(1.248,1.681,1.74)	1.618
ES_{112}	(2,8,4,4,6)	(0.62,0.47,0.32)	(1.736,2.068,1.92)	1.988
ES_{113}	(4,5,6,7,2)	(0.45,0.35,0.25)	(1.8,1.96,1.8)	1.906
ES_{114}	(6,8,8,9)	(0.46,0.32,0.18)	(3.128,2.56,1.62)	2.498
ES_{115}	(3,4,5,6,6)	(0.31,0.2,0.1)	(1.054,1,0.66)	0.952*
ES_{116}	(1,2,2,3,3,5)	(0.58,0.44,0.3)	(0.696,1.012,1.05)	0.965*
ES_{117}	(3,5,4,7,5,9)	(0.59,0.44,0.3)	(2.065,2.068,1.77)	2.017
ES_{121}	(2,6,4,1,5,6)	(0.51,0.38,0.26)	(1.326,1.558,1.456)	1.502
ES_{122}	(2,3,5,5)	(0.63,0.5,0.38)	(1.26,1.75,1.9)	1.693
ES_{123}	(3,2,4,7,6,2)	(0.48,0.35,0.22)	(1.536,1.645,1.364)	1.58
ES_{124}	(2,2,3,5,5)	(0.58,0.44,0.3)	(1.276,1.54,1.5)	1.489*
ES_{125}	(3,5,4,7,5,9)	(0.72,0.56,0.4)	(2.52,2.632,2.36)	2.568
ES_{126}	(2,4,3,8,5,3)	(0.62,0.47,0.32)	(1.488,1.786,1.696)	1.721
ES_{127}	(6,4,7,7,8,8)	(0.48,0.35,0.22)	(3.072,2.695,1.936)	2.631
ES_{128}	(4,6,6,2,7,8)	(0.68,0.56,0.44)	(3.128,3.472,3.432)	3.408
ES_{129}	(5,1,6,5,7,8)	(0.51,0.41,0.32)	(2.601,2.665,2.496)	2.626
ES_{1210}	(4,6,6,2,7,8)	(0.27,0.17,0.08)	(1.242,1.054,0.624)	1.013*
ES_{1211}	(2,2,3,8,5,4)	(0.42,0.29,0.16)	(0.924,1.102,0.864)	1.032*
ES_{1212}	(3,4,1,5,3)	(0.6,0.44,0.28)	(1.8,1.804,1.484)	1.75
ES_{1213}	(2,3,5,5,1)	(0.58,0.44,0.3)	(1.16,1.54,1.53)	1.475*
ES_{211}	(4,1,5,6,7)	(0.59,0.47,0.36)	(2.419,2.632,2.52)	2.577
ES_{212}	(3,2,5,6,8)	(0.5,0.35,0.2)	(1.6,1.75,1.36)	1.66
ES_{213}	(3,9,5,3,6,6)	(0.64,0.5,0.36)	(2.496,2.65,2.376)	2.578
ES_{221}	(0,2,0,8,1,8)	(0.43,0.29,0.16)	(0.086,0.232,0.288)	0.217*
ES_{222}	(2,3,5,5,1)	(0.66,0.5,0.34)	(1.32,1.75,1.734)	1.675
ES_{223}	(3,8,5,6,7,4)	(0.47,0.35,0.24)	(1.786,1.96,1.776)	1.9
ES_{224}	(6,4,7,7,8,8)	(0.52,0.38,0.24)	(3.328,2.926,2.112)	2.857
ES_{225}	(2,2,3,5,4,9)	(0.58,0.47,0.36)	(1.276,1.645,1.764)	1.603
ES_{226}	(4,4,5,6,6,9)	(0.58,0.41,0.24)	(2.552,2.296,1.656)	2.232
ES_{227}	(3,2,5,6,8)	(0.76,0.59,0.42)	(2.432,2.95,2.856)	2.848
ES_{311}	(4,6,6,2,7,8)	(0.44,0.29,0.16)	(2.024,1.798,1.248)	1.744
ES_{312}	(2,3,5,5)	(0.54,0.38,0.22)	(1.08,1.33,1.1)	1.25*
ES_{313}	(3,1,4,4,5,7)	(0.64,0.47,0.3)	(1.984,2.068,1.71)	1.994
ES_{321}	(3,3,4,4,5,5)	(0.38,0.26,0.14)	(1.254,1.144,0.77)	1.1*
ES_{323}	(5,3,6,5,7,6)	(0.41,0.29,0.18)	(2.173,1.885,1.368)	1.846

Notes: *Weaker attributes

The environmental sustainability index has been computed as (3.85, 4.95, 6.33) and by matching linguistic label (Table 9) with minimum D value, the institution has been identified as “Environmentally Sustainable”. The pictorial representation of environmentally sustainable label is shown in Figure 2.

Step 6. Identification and analysis of obstacles for improvement. The fuzzy logic method is also used for identifying the weak attributes impeding environmental sustainability. This is carried out in two steps. The first step is determining the fuzzy performance importance index using equation (5), and the second step includes ranking the attributes using the centroid technique by applying equation (6). Table 10 exhibits all the ranking scores of the environmental sustainability attributes:

$$FPII_{ijk} = U_{ijk} \times R_{ijk} \quad (5)$$

Development
of assessment
framework

where $U_{ijk} = (1 \ 1 \ 1) - G_{ijk}$

$$\text{Ranking Score} = \frac{a + 4b + c}{6} \quad (6)$$

5. Results and discussion

The model proposed in this study is aligned with other models as the model encapsulates the vital attributes that can facilitate environmental sustainability in HEIs. The model addresses environmental sustainability attributes across the curriculum, research, campus operations and university–community engagements, essential for sensitising students on environmental issues. Even though the model focuses only on the environment dimension, it is comprehensive and easy to understand compared to other models. Moreover, unlike the other models, this framework helps colleges identify attributes that require improvement. The application of the fuzzy logic method served two purposes: firstly, the development of the FESI that computes the overall environmental sustainability level of the institute under review, and secondly, the identification of weak attributes (Vinodh and Aravindraj, 2015). The computed environmental sustainability index reveals that the college is environmentally sustainable. After the ranking scores of attributes were obtained, the institution's management was consulted to determine the threshold value. The rationale behind consulting the management for setting the benchmark is to know how high the institution is willing to elevate its environmental sustainability capability in the future. If the management is not ready to aim high, they are more likely to adopt a lower threshold, whereas the opposite is true if the management is committed to promoting environmental sustainability. The management decided on 1.50 as the threshold value. This threshold value helped in identifying the weak attributes. On application of the threshold value, nine out of 35 environmental sustainability attributes were found to be weak.

The college is committed to enhancing environmental consciousness among staff and students, and several initiatives have been implemented in this regard. This section examines some of the institution's most effective strategies for improving environmental sustainability on campus. The institution under review has scored highest in the attribute, landscaping of the campus with trees and plants that consume less water (ES_{128}). The vegetation on the campus reflects the efforts taken by the staff and students in keeping the campus green. It is a regular feature of the nature club of the institution to grow trees and plants around the campus, spread across 80 acres of land. The college has also taken a biodiversity plot on lease to grow medicinal plants and fruit trees. The park also has a fish pond for rearing fish. The trees in the campus keep the air pollution and carbon footprint low (ES_{127}). The use of sanitiser is yet another high-scoring attribute (ES_{224}). The use of sanitiser began as a result of the Covid-19 pandemic. In response to the Covid-19 protocol, the college prepared sanitisers for college use. Hand sanitisers were also given to local government hospitals. As a commitment to conserving the environment, the solid waste generated on campus is treated using the incinerator and shredding machine (ES_{117}). The plastic waste on campus is shredded and given to the concerned authorities. The shredded plastic is used in the construction of roads. The college also has bins for biodegradable and non-degradable waste disposal (ES_{129}). The college has a rainwater storage system with a capacity of ten lakh litres. The facility helps in water and energy conservation (ES_{114}). The science departments are committed to environmental research and projects. Creating

environmental awareness regarding environmental issues is on the priority list of the college. Stickers are placed around electrical switchboards and water faucets as a reminder to switch off/close them after use (*ES*₂₁₃). Seminars, awareness campaigns are conducted regularly (*ES*₂₁₂). The institute operates a dynamic National Service Scheme (NSS) and Unnat Bharat Abhiyan (UBA) cell that engages students in environmental and rural development activities. Other initiatives include student engagement in environmental projects and research (*ES*₂₂₆), purchasing and using energy-efficient instruments, treating harmful chemicals in science labs before discharge and treating the food waste from canteen and dormitory kitchens using biogas plants. Soon, the college plans to increase the capacity of solar panels and completely change over from filament lamps to LED bulbs and energy-efficient ceiling fans. The college also aims to conduct the green audit and plans to incorporate the green building concept in its upcoming construction projects on campus. Besides the strong attributes, the research has also identified nine weak attributes. The following table outlines the weak attributes the institute needs to address to enhance its environmental sustainability index score ([Table 11](#)).

6. Practical/managerial implications

The study has contributed to the existing assessment frameworks on environmental sustainability in HE. The conceptual framework developed can be applied to any HEI for measuring its environmental sustainability level. The framework is comprehensive and easy to understand. The conceptual model includes all the environment sustainable activities, including education, research, campus operations and community outreach. The framework allows for easy calculations and fixing of a threshold value by the management of the respective institution. Conducting longitudinal study in the same institution would enable the institution to improve its sustainability index till the desired level is achieved. Furthermore, the management with every level of assessment can tighten the threshold value. Computation of the FESI would help the college understand its current environmental sustainability level, and identifying the attributes that require improvement would enable institutions to reorganise their processes and practices to improve their overall environmental sustainability index.

7. Conclusion

HEIs are social institutions responsible for economic and social transformations. HEIs have been recognised as the critical drivers for achieving Sustainable Development Goals ([Mori Junior et al., 2019](#)). This responsibility has placed impetus on universities and colleges to incorporate sustainable initiatives across their practices and operations. HEIs hold the responsibility of producing responsible citizens. Sensitising students with environmental issues and inculcating a sense of responsibility towards the environment through education, research, participation in sustainable practices on campus and outreach would enable them to reflect on their values, beliefs and actions and help them lead a sustainable life. Incorporating sustainability can be challenging as the concept is multi-dimensional in nature ([Rodriguez-Andara et al., 2018](#)). The framework developed through this study provides HEIs with a set of enablers, criteria and attributes that can facilitate environmental sustainability assessment in HEIs. A regular assessment of the sustainability level would help the institutions track their current practices and explore ways of promoting environmental sustainability within the given constraints and the available resources. The application of the fuzzy logic method for assessing the environmental sustainability level of the HEI served two purposes: firstly, the development of the FESI that shows the overall environmental sustainability level of the institute, and secondly, the identification of weak

Weaker attributes	Suggestions
Investment in reverse osmosis (RO) plant (<i>ES</i> ₁₁₅)	• Installation of RO plant on campus for clean and safe drinking water • One-time installation and regular maintenance of RO plant would help college cut on expenses of installing water purifier's that do not serve the purpose of providing safe drinking water
Investment in sensor-based lighting (<i>ES</i> ₁₁₆)	• College should switch completely to LED lighting • Use of sensor lighting with features like motion sensor, daylight-responsive sensing can be explored as they are commercially available and help in energy saving
Reuse of water from labs in watering plants on campus (<i>ES</i> ₁₂₄)	• Setting waste effluent treatment system for directing waste water from labs in watering plants on campus • Departments with labs can set up a small botanical garden, and provision can be made for using the lab water in watering the plants
Ban of single-use plastic (<i>ES</i> ₁₂₁₀)	• Substituting plastic with eco-friendly products • Canteens and campus stores should be disallowed from selling drinking water and other beverages in pet bottles • Sensitising students about the hazards of use of plastic • Encouraging use of cloth bags and reusable paper bags • Providing safe drinking water facility to discourage staff and students bring water in plastic bottles • Banning the use of paper cup with plastic lining, instead use utensils that can be reused
Reduce use of paper on campus (<i>ES</i> ₁₂₁₁)	• By shifting to digital platforms for communication within the institution • Taking notes in digital form • Using both sides of the paper • Rather than circulating printed notices, announcements can be made • Adopting selective printing • Purchasing recycled paper • Creating awareness on how to reduce the use of paper
Mechanism for e-waste disposal (<i>ES</i> ₁₂₁₃)	• The e-waste generated in the college should first be repaired and reused • Regular maintenance of computers, instruments and accessories. • Regularly disposing e-waste to the authorised vendors

*(continued)***Table 11.**
Suggestions given on
weaker attributes for
improvement

Table 11.

Weaker attributes	Suggestions
	• Use of rechargeable batteries • E-waste recycling policy • Setting centre for e waste collecting and disposal
Use of public transportation by students and staff (<i>ES</i> ₂₂₁)	• Student identity cards allow students to avail concession and encourage them to use public transport • College should encourage staff and students to use alternative modes of transportation like bicycling and electric vehicles • Charging points for electric vehicles • Commuter reward programmes for faculty members using carpooling • Discourage staff and students from using personal vehicles by offering novel incentives
Student participation in community engagements (<i>ES</i> ₃₁₂)	• Including community engagements into the curriculum in degree programmes • Extra credits to students for participating in the NSS activities and other environmental initiatives on and off-campus • Certificates and recognitions to students for participating in environmental-related programmes and workshops • Developing community engagement as the institution's best practice/distinctiveness
Funding sustainable project (<i>ES</i> ₃₂₁)	• College should avail subsidy on solar panels provided by the government of India • Government honours HEIs with cash prizes and recognition for promoting eco-friendly processes and practices on- and off-campuses • Government funding for institutions is available for constructing eco-friendly buildings

attributes (Vinodh and Aravindraj, 2015). The FESI for the college under study is (3.58, 4.95, 6.33). Computation of Euclidean distance revealed the HEI under review is “environmentally sustainable”. The calculation of FPII of environmental sustainability attributes found nine attributes out of 35 attributes to be weak. Suggestions have been proposed to address the weak attributes.

7.1 Limitations and future direction

This study has the following limitations: The model for the environmental sustainability assessment was tested in one HEI. The sample size was small, and the study only concentrated on the environmental aspect of sustainability. Integrating the

social and economic dimensions of sustainability in future research would help explore additional enablers, criteria and attributes that can aid in an overall assessment of HEIs. The model could be used to conduct longitudinal studies and comparative analysis between HEIs. Future studies could widen the scope of the conceptual model's use by involving more participants from different universities. Researchers in future can explore using trapezoidal or Gaussian membership functions to improve the effectiveness of outcomes.

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